

Explanations for Design Choices

- $|\varepsilon| = |S| \sin(\omega t + \phi + \arg S)$
 $|\varepsilon| < \frac{1}{1000} \implies$ we impose $|S| < \frac{1}{1000} \implies |W_S| > 1000$, for $\omega \in [0, 0.1]$
 $1000 = 60$ dB. I chose the following after a few tests:

$$W_S = \frac{2100}{\left(\frac{s}{0.1} + 1\right)^2}$$

The denominator is squared because without it, W_S did not decrease fast enough $\implies$ it would intersect W_T above 1. Because it is squared $\implies$ a 6dB error is introduced (66dB $\approx$ 2000, so I used 2100 as a safety margin).

- To construct the open-loop transfer function L , I wanted freedom of choice, so I placed $(100s+1)(0.1s+1)$ in the numerator of the compensator C to cancel out the left half-plane poles of the plant P .

So, for now, I have $L = \frac{10}{s}$.

From the graphical conditions $\implies L(0) > 67$ dB, $L(0.1) > 61$ dB, and $L(100) < -0.3$ dB. $L(0) > 67$ was automatically ensured by the pole at 0, but $L(0.1) < 61$ with this initial compensator.

I decided to increase the gain of L with an element $C(s) = K = 12 \implies L = \frac{120}{s}$. This easily surpassed the performance condition, but the gain was too high for the stability condition. Thus, I introduced a pole at 100 $\implies L = \frac{120}{s\left(\frac{s}{100} + 1\right)}$. It fit the constraints perfectly, but $C = \frac{12(100s+1)(0.1s+1)}{\frac{s}{100}+1}$ is unacceptable because it is not a proper transfer function. It must be made proper, so I added 2 poles that would not affect the performance or stability: $\omega_1 = 1000$, $\omega_2 = 10000$.

$$\implies L = \frac{120}{s \left(\frac{s}{100} + 1\right) \left(\frac{s}{1000} + 1\right) \left(\frac{s}{10000} + 1\right)}$$

- With this L , I noticed a spike at 10^2 in the $|W_S S| + |W_T T|$ plot.
 $\implies L$ had to be decreased $\implies$ required a phase lag compensator, which decreases the loop gain at frequencies higher than b .

$$C(s) = \frac{b}{a} \cdot \frac{s+a}{s+b}$$

Since $\omega = \sqrt{ab}$, I chose $\omega = 1$ (L behaved very well up to $\omega = 1$), and the easiest numerical values to test were $a = 2$ and $b = 0.5$ (which also worked perfectly) $\implies C(s) = \frac{0.5}{2} \cdot \frac{s+2}{s+0.5}$.

$$\implies L(s) = \frac{0.5}{2} \cdot \frac{s+2}{s+0.5} \cdot \frac{120}{s \left(\frac{s}{100} + 1\right) \left(\frac{s}{1000} + 1\right) \left(\frac{s}{10000} + 1\right)}$$